A Reproducible, Pre-Registered Test of Mechanical Elliott-Wave Signals, and Whether a Local-LLM Scenario Weigher Improves Calibration

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Working paper — deterministic and local-LLM arms complete, with statistics

Abstract

We build a deterministic, point-in-time Elliott-Wave (EW) signal generator and an independent reliability tester that share one set of outcome rules, and ask (1) whether the mechanical engine yields a tradeable, calibrated edge, and (2) whether a local-LLM scenario weigher improves it. On an anonymized 50-instrument panel (1962–2026; 588k daily bars), a monthly-cadence walk-forward produced 442 out-of-sample setups: 22.6% win rate (Wilson 95% CI 19.0–26.8%) and –0.01R expectancy (bootstrap 95% CI –0.17 to +0.15; p≈0.83) — a precise zero. (A smaller annual run had shown +0.35R; 6.5× the sample dissolved it, i.e. it was drift/noise.) The result is market drift, not wave skill: long setups profit (+0.36R), shorts lose (–0.50R), an all-long book matches the engine, and against buy-and-hold the engine's alpha is significantly negative (–6.06R, 95% CI –6.94 to –5.26). Two controls confirm this is not a mechanization artifact: tuning the most powerful parameter created an edge that vanished once both the parameter and the instruments were held out (p=0.59), and a gradient-boosted weigher's apparent skill was entirely the direction sign (drift), wave-structure features at chance (AUC 0.53). One positive survives, under the LLM weigher: its grade-A setups beat grade-B by ~0.5R (label-permutation p=0.001; +8pp win rate, p=0.04) — absent under the deterministic grade (p=0.11), robust to stock clustering, running against a reward/risk confound, and partially surviving drift adjustment (p=0.07); but it is relative ranking, not edge (overall expectancy is zero, grade-A still loses to buy-and-hold, and it is uneven across ticker halves). Confidence is poorly calibrated as a probability (Brier worse than base rate; ECE≈0.35; systematic overconfidence) though it discriminates. We report a reproducible null on profitability, with a fragile but real LLM ranking signal that warrants more data.

1. Introduction

EW analysis is popular but hard to automate objectively; critics cite its subjectivity and the ease of fitting counts in hindsight [1,2]. We sidestep the 'one true count' by emitting ranked candidate counts with confidence, and ask the only questions that matter for use: are the signals reliable out-of-sample, is the confidence calibrated, and does the engine beat the trivial benchmark of simply holding the market? A strict generator/tester split keeps the measured system from measuring itself.

2. Related work — how prior work ranks EW

The academic mainstream is skeptical of technical analysis in general and EW in particular. Lo, Mamaysky & Wang [3] built the reference automated framework — kernel-regression pattern recognition on U.S. stocks 1962–1996 — and found only modest incremental information in some patterns. Sullivan, Timmermann & White [4], applying White's Reality Check bootstrap to ~100 years of the DJIA (extending Brock, Lakonishok & LeBaron [5]), found that once data-snooping across the rule universe is corrected, out-of-sample profitability largely vanishes. EW-specific studies are fewer and mixed: D'Angelo & Grimaldi [6] report accurate EUR/USD forecasts, and regional studies [7,8] claim support — but these are largely in-sample, single-market, or analyst-discretion fits. The recurring critique is the same two problems: subjectivity of the count and data-snooping / hindsight.

What we replicate. A multi-decade, multi-stock U.S. panel overlapping Lo et al.'s era and design philosophy, with the out-of-sample + bootstrap discipline of [4]. We attack the two standard critiques directly: subjectivity → a fully deterministic, golden-record-hashed engine; data-snooping/hindsight → anonymized instruments (no names/dates/prices reach the analyst or LLM), a hard no-lookahead boundary, a disjoint-selection test of our main parameter (§5.1), and pre-registration of the LLM arm.

What we do not (yet) replicate. We test tradeable setups, not Lo et al.'s conditional-return-distribution estimand. We do not yet run White's Reality Check / Hansen's SPA across a full rule universe [4]. And unlike the positive regional EW studies [6–8], we forbid in-sample fitting and analyst-chosen counts — which plausibly explains why our out-of-sample result is weaker than their in-sample claims.

3. Methods

3.1 Data. 50 anonymized instruments (stock_id only), 588,427 daily OHLCV bars, 1962–2026, positive prices, no gaps.

3.2 Engine. Log-aware ZigZag pivots → bounded, beam-pruned, bidirectional sweep over motive (impulse, diagonals) and corrective (zigzag, flat, triangle) structures → scale-aware cardinal rules (W2<W1; W3 not shortest; W4 overlap) + guideline scores (Fibonacci bands, alternation, equality, volume) → three-degree nesting (daily ⊂ weekly ⊂ monthly) → weighted scenarios + residual bucket → R/R-gated setup. Large moves judged on log scale.

3.3 Setup construction. Entry/stop anchored to the completion pivot; reward/risk on log scale; degree-scaled invalidation, stop, and evaluation horizon; stable ids to de-duplicate one structure re-seen across bars.

3.4 No-lookahead & determinism. A signal at T uses only bars ≤ T (as_of clamp + tripwire test); deterministic, golden-record hashed; frozen JSON carries data hash + engine version.

3.5 Tester & statistics. An independent scorer resolves each setup against the continuation (won/lost/invalidated/expired; stop-first tie-break) and reports: win rate (Wilson 95%); expectancy in R (bootstrap 95%, and block-bootstrap by instrument for clustering); a random-direction permutation null; a drift benchmark (buy-and-hold from entry, and every setup forced long); calibration by grade and confidence, a Brier score and expected calibration error (ECE) against realized wins; a label-permutation test of any grade effect with a disjoint-ticker (split-half) replication; and transaction-cost sensitivity.

4. Results — deterministic engine

27,644 point-in-time signal records across the 50 instruments (monthly walk-forward, full history) yielded 442 distinct setups. A smaller annual run gave 68 setups at +0.35R; that positive dissolved to zero under 6.5× the sample and is reported here as the drift artifact it was.

Metric	Value	Inference
Win rate	22.6%	Wilson95 [19.0, 26.8]%
Expectancy (raw)	-0.01 R/trade	boot95 [-0.17, +0.15]; block-boot by stock [-0.25, +0.25]; p=0.83
Random-direction null	-0.11 R	5-95% [-0.21, -0.01]; p(eng≥rand)=0.06 (drift, see below)
Grade A / Grade B	23% win, +0.16 R (n=115)	23% win, -0.07 R (n=327)
A beats B? / calibrated?	No (perm p=0.11, n.s.)	No (Brier worse than base rate)

Drift control (the decisive test). The direction-randomizing null does not randomize market drift, so a mostly-long book in a rising market beats it by construction. Benchmarking against simply holding the market removes the illusion:

Benchmark	Mean R	Reading
Long setups	n=249, +0.36 R	all the profit is here
Short setups	n=193, -0.50 R	they lose
Force every setup long	+0.32 R	matches the engine (-0.01R)
Buy-and-hold from entry	+6.05 R	the actual drift
Engine alpha vs buy-and-hold	-6.06 R	boot95 [-6.94, -5.26]

Reading. Expectancy is a precise zero and the engine loses to buy-and-hold by –6.06R: profit comes only from long setups, the short calls lose, and an all-long book matches the engine, so its directional choices add nothing over drift. The deterministic grade hints at calibration (A +0.16R vs B –0.07R) but the separation is not significant (label-permutation p=0.11; block-bootstrap CI by stock includes zero; identical A/B win rates) and reverses after drift adjustment — so the engine's own grade carries no reliable ranking skill. Net: no wave-counting edge and no significant calibration from the deterministic engine; whether the weighting layer adds ranking information is tested in §7.

5. Robustness: can tuning or a learned weigher create an edge?

5.1 The pivot-sensitivity switch. The engine's highest-leverage free parameter is the pivot reversal threshold. We replaced the fixed-percentage threshold with a structural one — threshold = ln(1 + k·ATR/close) on each timeframe's own bars — so a single coefficient k adapts sensitivity to each instrument's volatility, and asked whether tuning k manufactures an edge (direction scored against a geometry-matched random-side null). Selecting k by looking at a window makes that window glow: k=5 beat its null at p≈0.03 and even replicated on a time-held-out split (p≈0.02). The decisive test holds out both the parameter and the instruments — one random half of the tickers chose k (it picked k=4), the other half was scored once:

Held-out test	n	Expectancy vs random-side null (R)
k=5, selected on this window	28	obs +0.640 / null +0.041 (p=0.029)
k=5, time-held-out (pre-2015)	52	obs +0.777 / null +0.231 (p=0.018)
k=4, parameter + names held out	20	obs +0.072 / null +0.153 (p=0.591)

On the fully held-out half the observed expectancy falls below the geometry null (p=0.59) — no edge. The 'best' k is unstable (k=4 on one half, k=5 on the panel holdout) and the obs-vs-null gap collapses once selection is forbidden — reproducing, inside our own pipeline, the central finding of [4].

5.2 A learned (gradient-boosted) weigher. We built a supervised table (one row per candidate scenario: the anonymized structural features of §3.2 plus its implied direction; label = whether that direction matched the sign of the forward 252-day return), trained a histogram GBT (depth 3, isotonic-calibrated) on one random half of the instruments, and evaluated once on the other.

Held-out predictor	AUC	Reading
GBT, all features	0.674	apparently predictive
Implied-direction sign alone	0.675	explains all of it (drift)
GBT, structure only (no direction)	0.529	wave features ≈ chance
GBT, structure only, long setups	0.467	no separation within a side

AUC 0.674 is not wave skill: the identical AUC (0.675) comes from the direction sign alone, because over 252-day windows the panel drifts up (P(correct | long)=0.70 vs 0.35 for short). Removing direction, wave structure alone gives AUC 0.529 (0.467 within long setups) — chance (label-shuffled null 0.502). The structural features carry no out-of-sample directional information beyond drift; a learned weigher over them can only follow drift, which §4 already shows loses to buy-and-hold.

6. LLM scenario weigher — pre-registration

A drop-in weigher (local model via Ollama, qwen2.5:7b-instruct, temperature 0, responses cached) re-weights the engine's existing candidate scenarios (and may change lead direction and confidence), leaving pivots/rules/structure/setup untouched. Controls: determinism (temp 0 + cache); no leakage (the model sees only anonymized relative features — structure, degree, leg proportions, Fibonacci fit, price position); identical universe, tester, horizon. Pre-registered success: expectancy CI clearing zero, grade A > B, monotonic confidence–win relationship, and the engine escaping the drift benchmark.

7. Results — LLM weigher

Metric	Deterministic	LLM (qwen2.5:7b-instruct)
Setups resolved	442	466
Win rate	22.6%	22.7%
Expectancy	-0.01 R [-0.17, +0.15]	+0.01 R [-0.14, +0.18], p=0.89
Grade A / B (meanR)	+0.16 (n=115) / -0.07 (n=327)	+0.32 (n=155) / -0.14 (n=311)
Grade A > B?	No (perm p=0.11)	Yes (perm p=0.001)

The grade signal. The LLM weigher's grade separation is the one result that survives scrutiny. Grade-A setups beat grade-B by ~0.5R (label-permutation p=0.001); the gap survives block-bootstrapping by stock (95% CI [+0.12, +1.01]) and appears in the win rate too (27.0% vs 19.1%, +8pp, p=0.04) — which runs against a reward/risk confound, since grade-A setups carry higher R:R (3.0 vs 2.4) and should therefore win less, not more. It partially survives drift adjustment (gap +1.19R after subtracting each setup's buy-and-hold return, p=0.07). The same test on the deterministic grade is null (p=0.11). So the LLM adds genuine relative ranking information, not merely concentrated drift.

Split-half replication. Splitting the 50 tickers into disjoint halves, the grade A>B expectancy gap holds on both (even +0.71R, p=0.001; odd +0.42R, p=0.05), but the win-rate edge concentrates in one half (even +15pp, p=0.005; odd +2pp, p=0.46). So the effect is directionally reproducible but uneven — a real but fragile signal, not yet a stable, generalizable edge.

Calibration (Brier / ECE). As a probability the confidence is poorly calibrated: Brier 0.33 (LLM) and 0.31 (deterministic), both worse than a base-rate forecast (~0.17–0.18); ECE≈0.35–0.39. It is systematically overconfident — the top bucket states ~82% but wins ~30%:

Confidence bucket	Stated	Actual win	n
70–100%	0.82	0.31	162
50–70%	0.58	0.18	132
30–50%	0.41	0.15	142
0–30%	0.27	0.00	13

Win rate rises with confidence (discrimination), but the levels are far above realized wins (miscalibration). Part of this is structural — the confidence expresses belief in the count's direction, scored here against a high-R:R win bar — but a user should treat the percentages as an ordering, not a probability. **Verdict.** The LLM weigher does not create an edge (expectancy zero; grade-A still loses to buy-and-hold), but it does carry a real, if fragile, ranking signal the deterministic engine lacks — the study's one positive, pending more data to confirm it is not a large-sample fluke.

8. Toward greater statistical power (future work)

The open question is now the LLM ranking signal, not the profitability null. Concrete steps: (a) finer (weekly) cadence and a universe beyond these 50 names — more setups to confirm or kill the grade-A effect, which is uneven across ticker halves; (b) tag the model in every record (currently only the weigher name is stored) so the LLM arm is provably a single model; (c) recalibrate confidence to a true win-probability (isotonic on a training fold) and re-measure Brier/ECE; (d) White's Reality Check / Hansen's SPA across configurations to bound data-snooping [4]; (e) a volatility-matched random-entry benchmark and transaction-cost curves (partially implemented). Pre-registration (done for the LLM arm) constrains researcher degrees of freedom.

9. Limitations

Monthly cadence still misses sub-monthly setups (a weekly run is the next step); finer cadence also produces clustered, semi-redundant setups, so raw n overstates independent information and the block-bootstrap-by-stock interval is the honest one. One engine family with un-tuned secondary parameters (beam width, R/R floor, horizon scaling); frictionless, point-in-time; one anonymized 50-instrument universe; a single local LLM at temperature 0. The grade-A finding is not multiple-testing corrected across the several metrics examined. Research artifact, not trading advice.

10. Conclusion

A reproducible, pre-registered harness gives a precise answer on this universe: the mechanical engine has no statistically detectable out-of-sample edge and no wave-counting skill. Expectancy is a precise zero on 442 monthly setups (a smaller annual run's +0.35R dissolved under 6.5× the data), and the engine underperforms buy-and-hold by –6.06R. The deterministic grade shows no significant calibration (p=0.11); tuning the most powerful parameter produced an edge that vanished once parameter and instruments were both held out; and a gradient-boosted weigher's skill was drift, not structure. The one positive is the local-LLM weigher's grade separation (A>B, p=0.001) — robust to clustering, running against a reward/risk confound, partially surviving drift adjustment — but it is relative ranking, not edge (expectancy zero; grade-A still loses to buy-and-hold) and it is uneven across ticker halves, so it warrants more data before any claim. The bottleneck is the information in the wave-count features, not the weighting model — reproducing, inside our own pipeline, the data-snooping and no-edge cautions of the technical-analysis literature [3,4].

References

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Artifacts: engine ewt/; testers tester.py, aggregate_tester.py; disjoint-selection sweep_sensitivity.py (--pivot-mode atr); weigher ewt/weigh/; runners batch_universe.py, weighed_walkforward.py; protocol EXPERIMENT.md, RUN_LLM_ARM.md.